

1.6 More on Linear Systems and Invertible Matrices

This section shows how the inverse of a matrix can be used to solve a linear system, and develops further results about invertible matrices.

Number of Solutions of a Linear System

Section 1.1 asserted (from Figures 1.1.1, 1.1.2) that every linear system has zero, one, or infinitely many solutions. We can now prove it.

Theorem 1.6.1

A system of linear equations has zero, one, or infinitely many solutions. There are no other possibilities.

Proof For $A\mathbf{x} = \mathbf{b}$, exactly one of “no solutions,” “exactly one,” or “more than one” holds; it suffices to show the last case gives infinitely many. Suppose $\mathbf{x}_1, \mathbf{x}_2$ are distinct solutions and let $\mathbf{x}_0 = \mathbf{x}_1 - \mathbf{x}_2 \neq \mathbf{0}$. Then $A\mathbf{x}_0 = A\mathbf{x}_1 - A\mathbf{x}_2 = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so for any scalar k ,

$$A(\mathbf{x}_1 + k\mathbf{x}_0) = A\mathbf{x}_1 + k(A\mathbf{x}_0) = \mathbf{b} + k\mathbf{0} = \mathbf{b}$$

Since $\mathbf{x}_0 \neq \mathbf{0}$ and k ranges over infinitely many scalars, $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. ■

Solving Linear Systems by Matrix Inversion

Beyond Gauss–Jordan and Gaussian elimination, invertibility gives an explicit *formula* for the solution.

Theorem 1.6.2

If A is an invertible $n \times n$ matrix, then for every $n \times 1$ matrix $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has exactly one solution: $\mathbf{x} = A^{-1}\mathbf{b}$.

Proof $A(A^{-1}\mathbf{b}) = \mathbf{b}$ shows $\mathbf{x} = A^{-1}\mathbf{b}$ is a solution. If $\mathbf{x}_0$ is any solution, then $A\mathbf{x}_0 = \mathbf{b}$, and multiplying by A^{-1} gives $\mathbf{x}_0 = A^{-1}\mathbf{b}$ – so it’s the only one. ■

EXAMPLE 1 | Solution of a Linear System Using A^{-1}

Solve $x_1 + 2x_2 + 3x_3 = 5$, $2x_1 + 5x_2 + 3x_3 = 3$, $x_1 + 8x_3 = 17$.

SOLUTION: In matrix form $A\mathbf{x} = \mathbf{b}$ with $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$. From Example 4 of Section 1.5,

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}, \text{ so}$$

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

i.e., $x_1 = 1, x_2 = -1, x_3 = 2$. (This method only applies when the system is square and the coefficient matrix is invertible.)

Linear Systems with a Common Coefficient Matrix

For a sequence $A\mathbf{x} = \mathbf{b}_1, \dots, A\mathbf{x} = \mathbf{b}_k$ sharing an invertible A , the solutions $\mathbf{x}_i = A^{-1}\mathbf{b}_i$ need one inversion and k multiplications. More efficiently, form

$$[A \mid \mathbf{b}_1 \mid \mathbf{b}_2 \mid \cdots \mid \mathbf{b}_k] \quad (1)$$

and reduce to reduced row echelon form by Gauss–Jordan elimination, solving all k systems at once – and this works even when A is not invertible.

EXAMPLE 2 | Solving Two Linear Systems at Once

Solve

(a) $x_1 + 2x_2 + 3x_3 = 4, 2x_1 + 5x_2 + 3x_3 = 5, x_1 + 8x_3 = 9$

(b) $x_1 + 2x_2 + 3x_3 = 1, 2x_1 + 5x_2 + 3x_3 = 6, x_1 + 8x_3 = -6$.

SOLUTION: Both share the coefficient matrix of Example 1. Augmenting with both constant columns:

$$\left[\begin{array}{ccc|cc} 1 & 2 & 3 & 4 & 1 \\ 2 & 5 & 3 & 5 & 6 \\ 1 & 0 & 8 & 9 & -6 \end{array} \right] \xrightarrow{\text{reduce to RREF}} \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \end{array} \right]$$

The last two columns give the solutions:

(a) $x_1 = 1, x_2 = 0, x_3 = 1$;

(b) $x_1 = 2, x_2 = 1, x_3 = -1$.

Properties of Invertible Matrices

To show A is invertible has meant finding B with $AB = BA = I$. In fact, either equation alone forces the other.

Theorem 1.6.3

Let A be square.

(a) If $BA = I$ for square B , then $B = A^{-1}$.

(b) If $AB = I$ for square B , then $B = A^{-1}$.

Proof (a) It suffices to show A is invertible; then multiplying $BA = I$ on the right by A^{-1} gives $B = A^{-1}$.

By Theorem 1.5.3, it suffices to show $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. If $A\mathbf{x}_0 = \mathbf{0}$, multiplying on the left by B gives $BA\mathbf{x}_0 = B\mathbf{0}$, i.e., $\mathbf{x}_0 = \mathbf{0}$. ■

Equivalence Theorem

We can now add two more statements to Theorem 1.5.3.

Theorem 1.6.4 – Equivalent Statements

For an $n \times n$ matrix A , the following are equivalent:

- (a) A is invertible.
- (b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (c) The reduced row echelon form of A is I_n .
- (d) A is a product of elementary matrices.
- (e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.
- (f) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $n \times 1$ matrix $\mathbf{b}$.

Proof Since (a)–(d) are already equivalent (Theorem 1.5.3), it suffices to show $(a) \Rightarrow (f) \Rightarrow (e) \Rightarrow (a)$.

$(a) \Rightarrow (f)$: This is Theorem 1.6.2.

$(f) \Rightarrow (e)$: Immediate, since having exactly one solution implies consistency.

$(e) \Rightarrow (a)$: Apply consistency to each of the systems $A\mathbf{x} = \mathbf{e}_1, \dots, A\mathbf{x} = \mathbf{e}_n$ (where $\mathbf{e}_i$ is the i th standard unit column). Taking solutions $\mathbf{x}_1, \dots, \mathbf{x}_n$ as columns of a matrix $C = [\mathbf{x}_1 \mid \dots \mid \mathbf{x}_n]$, then $AC = [A\mathbf{x}_1 \mid \dots \mid A\mathbf{x}_n] = [\mathbf{e}_1 \mid \dots \mid \mathbf{e}_n] = I$. By Theorem 1.6.3(b), $C = A^{-1}$, so A is invertible. ■

We know invertible factors give an invertible product; the converse also holds.

Theorem 1.6.5

Let A, B be square of the same size. If AB is invertible, then A and B are both invertible.

Proof First, B is invertible: if $B\mathbf{x}_0 = \mathbf{0}$ for some $\mathbf{x}_0$, then $(AB)\mathbf{x}_0 = A(B\mathbf{x}_0) = A\mathbf{0} = \mathbf{0}$; since AB is invertible, Theorem 1.6.4 forces $\mathbf{x}_0 = \mathbf{0}$, so $B\mathbf{x} = \mathbf{0}$ has only the trivial solution and B is invertible. Then $A = A(BB^{-1}) = (AB)B^{-1}$ is a product of invertible matrices, hence invertible. ■

A fundamental question recurs throughout the text:

A Fundamental Problem

For a fixed $m \times n$ matrix A , find all $m \times 1$ matrices $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ is consistent.

If A is invertible, Theorem 1.6.2 fully answers this: every $\mathbf{b}$ works. Otherwise (non-square, or square but singular), $\mathbf{b}$ must usually satisfy some condition, found as in the next examples.

EXAMPLE 3 | Determining Consistency by Elimination

What must b_1, b_2, b_3 satisfy for $x_1 + x_2 + 2x_3 = b_1$, $x_1 + x_3 = b_2$, $2x_1 + x_2 + 3x_3 = b_3$ to be consistent?

SOLUTION: Row-reducing the augmented matrix $\begin{bmatrix} 1 & 1 & 2 & b_1 \\ 1 & 0 & 1 & b_2 \\ 2 & 1 & 3 & b_3 \end{bmatrix}$ (add $-1 \times$ row 1 to row 2 and

$-2 \times \text{row 1}$ to row 3; multiply row 2 by -1 ; add row 2 to row 3) yields

$$\begin{bmatrix} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_1 - b_2 \\ 0 & 0 & 0 & b_3 - b_2 - b_1 \end{bmatrix}$$

so the system is consistent iff $b_3 - b_2 - b_1 = 0$, i.e., $b_3 = b_1 + b_2$ – that is, iff $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_1 + b_2 \end{bmatrix}$ for arbitrary b_1, b_2 .

EXAMPLE 4 | Determining Consistency by Elimination

What must b_1, b_2, b_3 satisfy for $x_1 + 2x_2 + 3x_3 = b_1$, $2x_1 + 5x_2 + 3x_3 = b_2$, $x_1 + 8x_3 = b_3$ to be consistent?

SOLUTION: Reducing $\begin{bmatrix} 1 & 2 & 3 & b_1 \\ 2 & 5 & 3 & b_2 \\ 1 & 0 & 8 & b_3 \end{bmatrix}$ to reduced row echelon form gives

$$\begin{bmatrix} 1 & 0 & 0 & -40b_1 + 16b_2 + 9b_3 \\ 0 & 1 & 0 & 13b_1 - 5b_2 - 3b_3 \\ 0 & 0 & 1 & 5b_1 - 2b_2 - b_3 \end{bmatrix} \quad (2)$$

Here b_1, b_2, b_3 are unrestricted – the system is always consistent, with unique solution

$$x_1 = -40b_1 + 16b_2 + 9b_3, \quad x_2 = 13b_1 - 5b_2 - 3b_3, \quad x_3 = 5b_1 - 2b_2 - b_3 \quad (3)$$

for all b_1, b_2, b_3 . (This coefficient matrix is precisely the invertible A of Example 1 – consistent with Theorem 1.6.4(e).)

Exercise Set 1.6

1. Solve by inverting the coefficient matrix and using Theorem 1.6.2.

$$x_1 + x_2 = 2, \quad 5x_1 + 6x_2 = 9$$

3. $x_1 + 3x_2 + x_3 = 4$, $2x_1 + 2x_2 + x_3 = -1$, $2x_1 + 3x_2 + x_3 = 3$

6. $-x - 2y - 3z = 0$, $w + x + 4y + 4z = 7$, $w + 3x + 7y + 9z = 4$, $-w - 2x - 4y - 6z = 6$

7. Solve for x_1, x_2 in terms of b_1, b_2 : $3x_1 + 5x_2 = b_1$, $x_1 + 2x_2 = b_2$

9. Solve the linear systems together by reducing an appropriate augmented matrix to reduced row echelon form.

$$x_1 - 5x_2 = b_1, \quad 3x_1 + 2x_2 = b_2$$

$$(i) b_1 = 1, b_2 = 4 \quad (ii) b_1 = -2, b_2 = 5$$

13. Determine conditions on the b_i 's, if any, for the system to be consistent.

$$x_1 + 3x_2 = b_1, \quad -2x_1 + x_2 = b_2$$

15. $x_1 - 2x_2 + 5x_3 = b_1$, $4x_1 - 5x_2 + 8x_3 = b_2$, $-3x_1 + 3x_2 - 3x_3 = b_3$

18. For $A = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & -2 \\ 3 & 1 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$:

(a) Show that $A\mathbf{x} = \mathbf{x}$ can be rewritten as $(A - I)\mathbf{x} = \mathbf{0}$, and use this to solve $A\mathbf{x} = \mathbf{x}$.

(b) Solve $A\mathbf{x} = 4\mathbf{x}$.

19. Solve the matrix equation for X .

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 0 \\ 0 & 2 & -1 \end{bmatrix} X = \begin{bmatrix} 2 & -1 & 5 & 7 & 8 \\ 4 & 0 & -3 & 0 & 1 \\ 3 & 5 & -7 & 2 & 1 \end{bmatrix}$$

21. **(Working with Proofs)** Let $A\mathbf{x} = \mathbf{0}$ be a homogeneous system of n equations in n unknowns with only the trivial solution. Prove that for any positive integer k , $A^k\mathbf{x} = \mathbf{0}$ also has only the trivial solution.

23. **(Working with Proofs)** Let $A\mathbf{x} = \mathbf{b}$ be any consistent system, and let $\mathbf{x}_1$ be a fixed solution. Prove that every solution can be written $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_0$ where $\mathbf{x}_0$ solves $A\mathbf{x} = \mathbf{0}$, and that every matrix of this form is a solution.

24. **(Working with Proofs)** Use part (a) of Theorem 1.6.3 to prove part (b).